IMPERIAL

Independent Study Option

Neuromorphic Computing for Decentralised Federated and Constrained Edge AI devices

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Agenda

1. Introduction
2. Neuromorphic Computing
3. Federated Learning
4. Decentralised Edge AI Devices
5. Experiment
6. Implementation & Results
7. Future Works

Introduction

Neuromorphic Computing for Decentralised Federated and Constrained Edge AI devices

The research aims to explore how **Neuromorphic Computing** can be **integrated** into the current **Decentralised Federated and Constrained Edge AI Devices**.

I started by Research by:

- Understanding **What** Neuromorphic Computing is.
- Familiarising myself with the **current developments** of Neuromorphic Computing.
- What **advantages** it offers?

What is Neuromorphic Computing?

“Neuromorphic computing covers a diverse range of approaches to **information processing** all of which demonstrate some degree of **neurobiological inspiration** that **differentiates** them from mainstream **conventional computing** systems.” - Steve Furber [1]

Specifically we are interested in emulating the Brain.

[1]https://iopscience.iop.org/article/10.1088/1741-2560/13/5/051001

How does the Brain process Information?

Current Neuroscience research believes that **neurons** and **synapses** plays a huge role in cognition.

Neurons are brain cells that functionally can be abstracted as a spiking function, while Synapses are biological structures that connect neurons.

Neuromorphic chips aims to **physically** emulate Neurons and Synapses.

Neuromorphic Computing Hardware Development Timeline

1952 :
Hodgkin-Huxley Neuron Model [1]

1971 :
Memristor (Theory) [2]

2008 :
Memristor (Physical Model) [3]

2014 :
IBM TrueNorth

2016 :
SpiNNaker
BrainScaleS

2017:
Intel Loihi 1
DYNAPs

2021 :
Loihi 2
SpiNNaker 2
Akida
Innatera

2022
Kapoho Point (Intel)

2024
Hala Point (Intel)

[1]https://pmc.ncbi.nlm.nih.gov/articles/PMC1392213/
[2]https://ieeexplore.ieee.org/document/1083337?utm_source=acs&getft_integrator=acs
[3]https://www.nature.com/articles/nature06932#citeas

Neuromorphic Devices
Characteristics

According to [1] Neuromorphic Computing have these characteristics

- Connectionism
- Asynchrony
- Parallelism
- Impulse nature of Information
- Sparsity
- Analogue Computing
- In-memory computing
- Continual learning with on-device learning

[1]https://www.frontiersin.org/journals/neuroscience/articles/10.3389/fnins.2022.959626/ful

Advantages :

1) Energy Efficient
2) Robustness
3) Decentralisation
4) Low Latency

Current State of Art
Loihi 2

- Max # of Synapses : 120 million
- Max # of Neurons : 1 million(~7.8x more than Loihi 1)
- Die Area : 31mm (~50% drop from Loihi 1)
- Shown to have satisfactory results by studies

[1]https://ieeexplore.ieee.org/abstract/document/10311158?figureId=fig7#fig7
[2]https://www.intel.com/content/www/us/en/research/neuromorphic-computing-loihi-2-technology-brief.html

Experiment by others :

1. Real-time Vision-based Control of SWaP-constrained Flight System with Intel Loihi 2 [1].
   a. Utilised Loihi 2 chips to learn how to pilot an Uncrewed Aerial System
   b. Typically done by GPUs, but hard to mount them on Aircraft.
   c. Loihi 2 was used, and it showed **exceptional and robust** performance.
2. Intel NRC Lab
   a. Intel NRC Lab members have evaluated Loihi chips on a wide variety of tasks
   b. In most of these tasks, Loihi consumed less than 1 W of power
   c. In some of these tasks, it achieved state-of-the-art response time, while learning continually

What about Software for Neuromorphic Computing?

- Due to the novelty of the field, software for Neuromorphic Computing is largely **fragmented** and **not standardised**
- This makes community uptake **challenging**, making it commercially inviable.
- Fortunately there has been efforts to standardise Software
- Together with the release of Loihi 2, Intel also released **Lava**

Lava

A Python framework that supports neuromorphic computing implementations.

Managed by Intel NRC Team and is Open-source.

Abstracts hardware complexity through a flexible programming framework that enables diverse implementations while shielding developers from l**ow-level technical details**.

It supports a **variety of neuromorphic devices** too!

- Key problem in fragmentation! Made it possible to **compare** devices!

In addition, it also provides **integration** to third-party frameworks like Pytorch, ROS and TensorFlow.

Learning in Neuromorphic Computing

- **Plasticity** is the variable for learning

- Architecture for Learning : **Spiking Neural Network**

- Algorithms for learning:
  - Surrogate Gradient Learning
  - STDP
  - Evolutionary Algorithms
  - Reservoir Computing

Spike Timing Dependent Plasticity

- A form of learning method induced by **tight temporal correlations** between the spikes of **presynaptic** and **postsynaptic** neurons [1]

- Idea : Neurons that **fire together wire together**

- Lava : Sum of Product
  - More **customisable** STDP Learning rule
- Lava :
  - Reward Modulated Learning Framework -> **Reinforcement Learning possible**

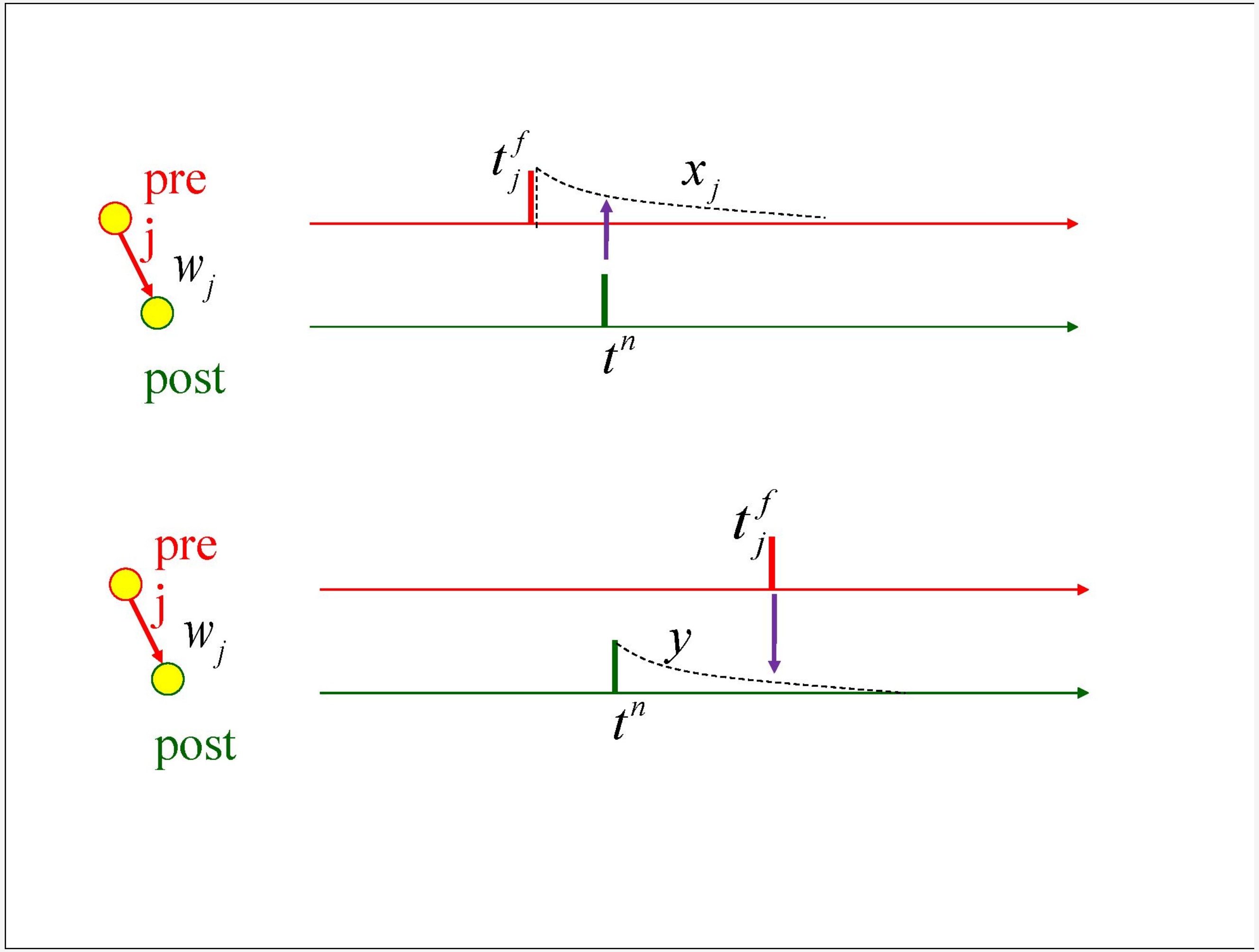


Image from http://www.scholarpedia.org/article/Spike-timing_dependent_plasticity

Federated Learning In Neuromorphic Computing

- Federated Learning: A learning framework that aims to ensure **data privacy** by **distributing learning**.

- Why Neuromorphic Computing:
  - Current frameworks are **vulnerable** to data transfer **bottlenecks** and **data interception** due to **reliance** on **central server**.
  - With Neuromorphic Computing, there might be **no need for a central server** since they potentially can **coordinate themselves**.
  - Furthermore, since computation is **distributed** in Neuromorphic Computing, it is **hard to recover the data** that has been diffused into the network.

Neuromorphic Computing for Decentralised Edge AI Devices

- Decentralised Edge devices usually refer to devices that have limited computation, and are deeply integrated into people's life
  - The push for edge devices is due to the **increased in data generated by users**.

- Why Neuromorphic Computing:
  - Although technological advances have made computation less scarce, these devices **still faces physical issues** like battery or size limitations.
  - This can **limit the impact of technological advances** in edge devices, since they physically cannot carry them.
  - However Neuromorphic Computing is proven to be more energy efficient, while providing robust results.
  - These devices might be able to **better capitalise** on Technological Advancement with Neuromorphic Computing

Experiment

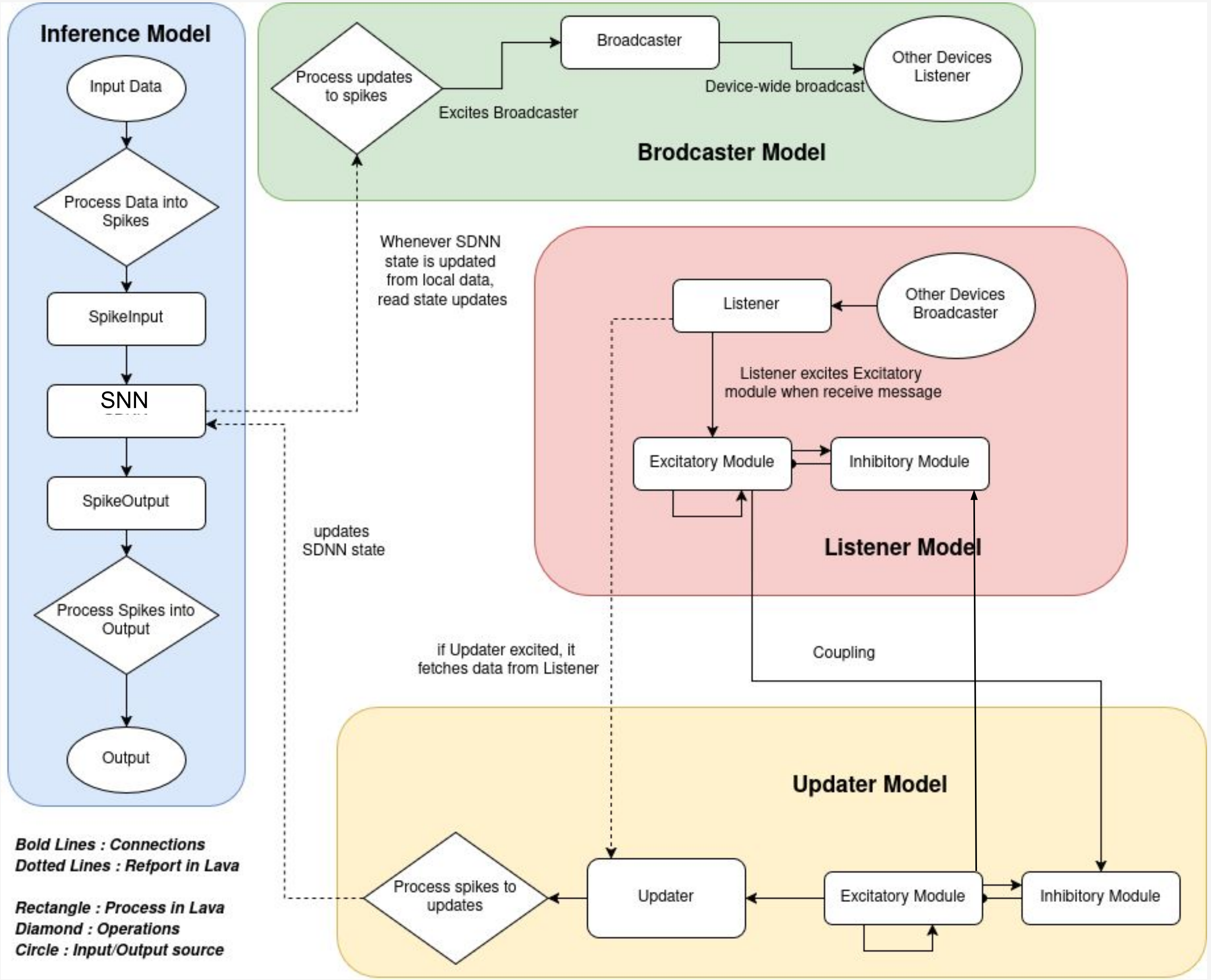
- Given an image with a person, determine whether he/she is wearing a mask
- **Binary Classification Task** where input is an image, and output is [0,1]
- Dataset from kaggle [1]
- For decentralised setup, I will use 4 raspberry Pis 3b with 1GB RAM and 8GB storage.
  - They are connected with HTTP connections
  - Federated Learning is applied through weight updates
- Used the lava main library only.

[1] https://www.kaggle.com/datasets/jishan900/covid-facemask-monitoring-dataset

Implementation

4 Entities

1. Inference Model
2. Broadcaster Model
3. Listener Model
4. Updater Model

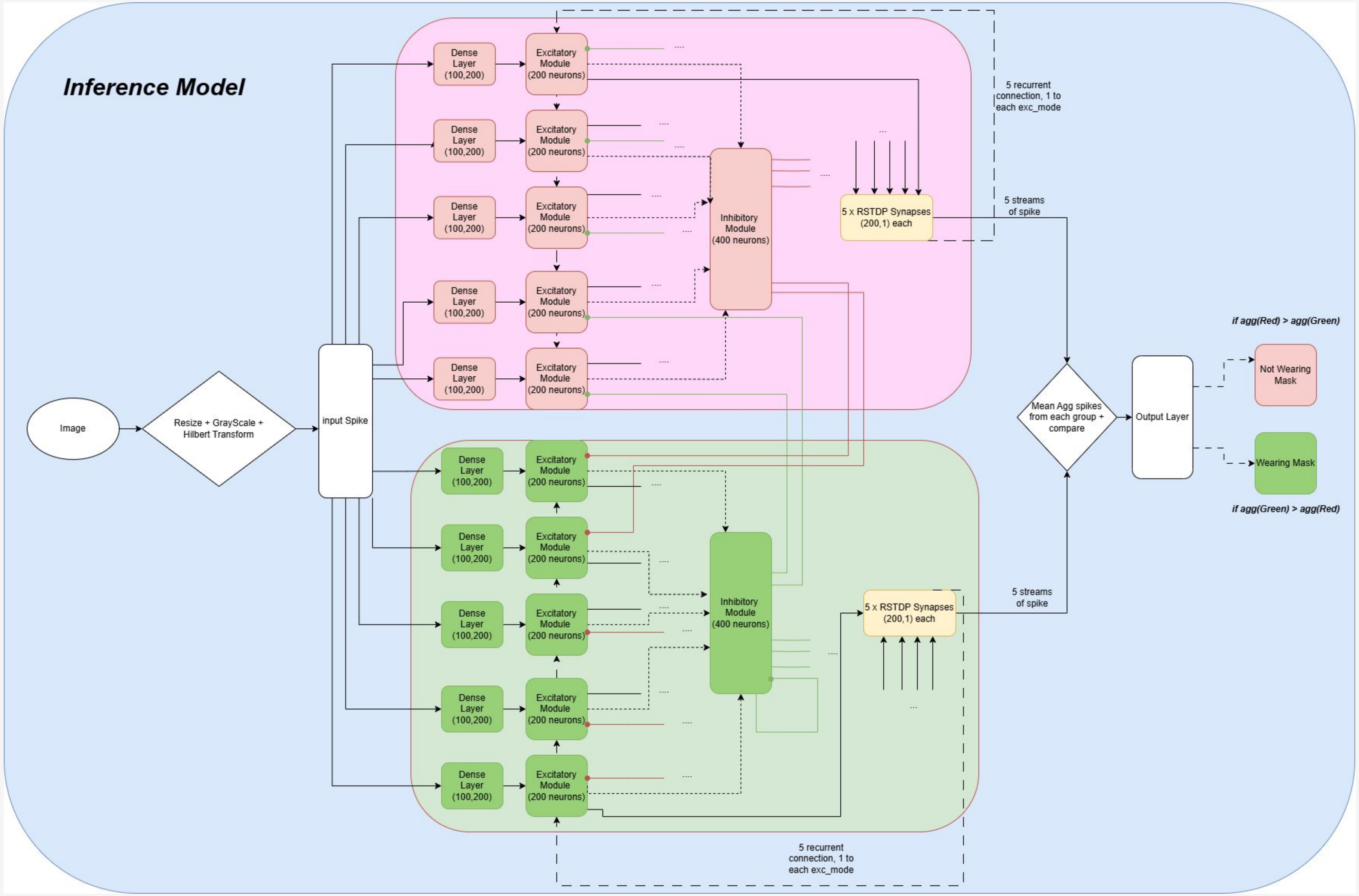


Implementation

Within the Inference Model:

1. 10 modules of excitatory neurons
2. 2 modules of Inhibitory neurons
3. 2 R-STDP dense layers

The neurons modules are split into 2 coalition in a **Winner-Take-All** competitive setup.



Implementation

$$STDP : \Delta w = \begin{cases} A_{+} e^{-\frac{\Delta t}{\tau_{+}}}, & \text{if } \Delta t > 0 \\ A_{-} e^{\frac{\Delta t}{\tau_{-}}}, & \text{if } \Delta t < 0 \end{cases}$$

$$\Delta w = M(t) \cdot STDP(\Delta t)$$

The equation above outlines how Spike-Timing-Dependent-Plasticity works

Variable A is basically the learning rate for pre- and postsynaptic.

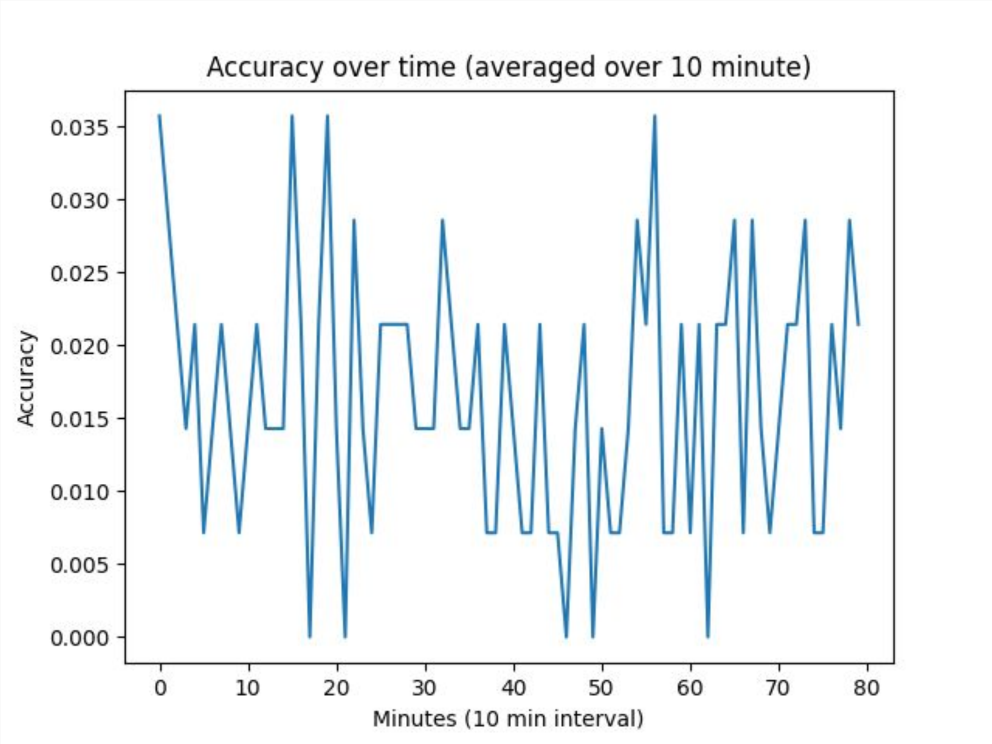
Since it is a Binary Classification task, and we have a WTA setup. It is possible to induce rewards into general STDP.

- Specifically we can **encourage 1 mega-module** of neurons while **discouraging** the other for each class label
- This can be done by alternating the signs of learning rate A

Results & Evaluation

- The Inference model **failed to run** on Raspberry PI and as such I could not continue with my decentralised setup.
    - As such, it **might not be ready** to be implemented in a Constrained Edge AI devices.

- I ran the model on my laptop, and it **failed to learn** even after 10 hours of training.
    - I severely suspect that either my **implementation was wrong**, or that the **network was not complex** enough to learn; or both.



Future Works

1. Explore more algorithms and methods that can be used to **train** SNN
2. Explore more **advance features** of Lava
3. Research how **learning** can be **transferred** between devices more effectively
    - Specifically from the point of view of **Information Theory** and **Graph Theory**

The End
Summary

Neuromorphic Computers

1. What they are?
2. Developments
3. Characteristics
4. Current State of Art

Learning in Neuromorphic Computers

1. Plasticity
2. Spiking Neural Network
3. Learning Algorithms
4. Federated Learning

Implementation

1. General Model
2. Inference Model
3. Learning in Inference Model.